Benjamin J. Linzmeier

Assistant Professor of Geology
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Employment

(August 2024 →) **University South Alabama Department of Earth Sciences**
Assistant Professor of Geology

(January 2024 – August 2024) **Uroboros Innovations, LLC**
Chief Scientist

(January – May 2024) **University South Alabama Department of Earth Sciences**
Part time instructor: Paleontology

(August 2021 – December 2023) **University South Alabama Department of Earth Sciences**
Assistant Professor of Geology

(October 2019 – July 2021) **University of Wisconsin - Madison**
WiscSIMS laboratory post-doctoral fellow

(August 2020 – December 2020) **University of Wisconsin - Madison**
Geoscience Department Lecturer

(2017 – 2019) **Northwestern University**
Ubben post-doctoral fellow
Ocean acidification in deep time

Education

(2009 – 2017) **University of Wisconsin–Madison**
Aug. 2017 **Ph.D.**, Department of Geosciences
advisors: S.E. Peters and J.W. Valley
May 2012 **M.S.**, Department of Geosciences

(2005 – 2009) **Bowling Green State University**
B.S., Geology, *magna cum laude*

Skills summary

laboratory: Cameca IMS-1280 ion microprobe isotope analyses, Cameca electron microprobe trace elements, scanning electron microscopy: cathodoluminescence, EBSD, and EDS. Thermal ionization mass spectrometery. X-ray fluorescence.
programming/computing: R; SQL; Inkscape; GIMP; QGIS; Large Language Models
teaching: Active learning; backwards-design; online learning.

2026 Sarathi A., Faulkner P., **Linzmeier, B. J.**, Ali A.K., Othman H., and Ussi A. (2026) Seasonal and gendered patterns in subsistence fishing and shellfish gathering: insights from Mfumbwi (Zanzibar) for archaeological interpretation and contemporary fisheries management. Frontiers in Environmental Archaeology 4:1732658. doi.org/10.3389/fearc.2025.1732658

2025 Schultze, S.R., #Martin, J., #West, K., and **Linzmeier, B. J.** Urban Microclimates in Action! High resolution temperature and humidity differences across diverse urban terrain. Atmosphere: doi.org/10.3390/atmos16040416

2022 **Linzmeier, B. J.**, Jacobson, A.D., Sageman, B.B., Hurtgen, M.T., Ankney, M.E. and Landman, N.H. Isotope systematics of subfossil, historical, and modern *Nautilus macromphalus* from New Caledonia. PLoS ONE: doi.org/10.1371/journal.pone.0277666

Hartman, S., Lovelace, D.M., **Linzmeier, B. J.**, Mathewson, P.D., and Porter, W.P. Mechanistic thermal modeling of Late Triassic terrestrial amniotes predicts biogeographic distribution. Diversity: doi.org/10.3390/d14110973

2021 Hoffman, R., **Linzmeier, B. J.**, Kitajima, K., Nehrke, G., Dietzel, M., Jöns, N., Stevens, K., and Immenhauser, A. Complex biomineralization pathways of the belemnite rostrum cause biased paleotemperature estimates. Minerals: doi.org/10.3390/min11121406

Hoffmann, R., Slattery, J.S., Kruta, I., **Linzmeier, B. J.**, Lemanis, R., Mironenko, A., Goolaerts, S., De Baets, K., Peterman, D.J., and Klug, C. Recent advances in heteromorph ammonoid palaeobiology. Biological Reviews: doi.org/10.1111/brv.12669

2020 **Linzmeier, B. J.**, Jacobson, A.D., Sageman, B.B., Hurtgen, M.T., Ankney, M.E., Petersen, S.V., Tobin, T.S., #Kitch, G.D., and #Wang, J. Calcium cycle instability before and during the end-Cretaceous extinction. Geology. doi.org/10.1130/G46431.1

Husson, J.M., **Linzmeier, B. J.**, Kitajima, K., Ishida, Akizumi, Peters, S.E., Maloof, A.C., Schoene, B., and Valley, J.W., Large isotopic variability at the micron-scale in 'Shuram' excursion carbonates from South Australia. Earth and Planetary Science Letters: doi.org/10.1016/j.epsl.2020.116211

Lovelace, D.M., #Hartman, S.H., #Mathewson, P., **Linzmeier, B. J.**, Krentz, D. and Porter, W., Modeling Dragons: Using linked mechanistic physiological and microclimate models to explore environmental, physiological, and morphological constraints on the early evolution of dinosaurs. PLoS One: doi.org/10.1371/journal.pone.0223872

2019 **Linzmeier, B. J.**, Refining the interpretation of oxygen isotope variability in free-swimming organisms. The Swiss Journal of Paleontology, vol. 138, p. 109-121.
PaleorXiv preprint: doi.org/10.31233/osf.io/nbtgm

2018 **Linzmeier, B. J.**, Landman, N. H., Peters, S. E., Kozdon, R., Kitajima, K., and Valley, J. W., Ion microprobe stable isotope evidence for ammonite habitat and life mode during early ontogeny: Paleobiology: doi.org/10.1017/pab.2018.21

Linzmeier, B. J. Kitajima, K., #Denny, A.C., and Cammack, J. N., Making maps on a micrometer scale: EOS, 99: doi.org/10.1029/2018EO099269

2016 **Linzmeier, B. J.**, Kozdon, R., Peters, S. E., and Valley, J. W., Oxygen isotope variability within *Nautilus* shell growth bands: PLoS ONE 11(4): `doi.org/10.1371/journal.pone.0153890`

Hoffman, R., Richter, D.K., Neuser, R.D., Jons, N., **Linzmeier, B. J.**, Lemanis, R.E., Fusseis, F., Xaio, X., and Immenhauser, A., Evidence for a composite organic-inorganic fabric of belemnite rostra: Implications for palaeoceanography and palaeoecology. Sedimentary Geology: `doi.org/10.1016/j.sedgeo.2016.06.001`

Papers in Preparation (# denotes student co-author)

Linzmeier, B. J., Bolstad, K., Kitajima, K., Landman, N.H., and Hoffman, R. Shallow-to-deepwater life cycle of the Giant Squid (*Architeuthis dux*). For submission to PNAS.

Linzmeier, B. J., Tobin, T.S., Ward, P.D., Orland, I.J., #Assing, D.A., Kitajima, K., Gopon, P., Huber, B.T., Peters, S.E., and Valley, J.W. High spatial resolution assessment for diagenesis and primary isotopic variability in Maastrichtian molluskan carbonates from Antarctica. Resubmit at PALAIOS.

Hartman, S.H., Lovelace, D.M., **Linzmeier, B. J.** Mathewson, P., and Porter, W. Cold-stress predicts End-Triassic Extinction of tetrapods. For submission to Geology.

Linzmeier, B. J., Sessa, J.A., Orland, I.J., Landman, N.H., Peters, S.E., and Valley, J.W. Planktic early ontogeny in Late Cretaceous Owl Creek scaphitid ammonites. For submission to Paleobiology.

Linzmeier, B. J., #Johnson, A.B., Husson, J.M., Leet, K., Ritterbush, K., Quinn, D.P., and Peters, S.E. Chert prevalence across Earth History. For submission to Geology.

Selected Professional Talks (* denotes invited talk; # denotes student co-author)

2025 ***Linzmeier, B.J.** Revealing paleobiology and paleoclimatology with 10 micrometer spots. UC-Davis departmental seminar.

***Linzmeier, B.J.**, Hoffmann, R., and Kitajima, K. Unpacking complex biomineralization pattern in belemnite rostra. GSA Connects, San Antonio, TX.

2024 Beebe, D.A., #Stanley, M. and **Linzmeier, B.J.**, An automated hydraulic potentiometer for measuring groundwater-surface water interactions. GSA Southeastern meeting.

Waldeck, A.R., Jacobson, A.D., Sageman, B.B., **Linzmeier, B.J.**, Kitch, G.D., Sarvian, N.L, #Wan, C., and Hurtgen, M.T. Radiogenic Sr isotope ratios spanning Cretaceous OAE2 record hydrothermal activity and ocean mixing. Goldschmidt.

2023 **Linzmeier, B.J.**, Bolstad, K., Kitajima, K., Landman, N.H., and Hart, M.B. Statolith geochemistry of modern and ancient cephalopods: A view from modern *Architeuthis dux* and specimens from the Callovian (Jurassic) Christian Malford Lagerstätte. GSA Connects.

#Iweka, R.C., #Foster, A.B., Smith, K.E., and **Linzmeier, B.J.** Soil Lead Contamination in Mobile, Alabama: Using Community Engagement to Address Environmental Justice. GSA Connects.

#Foster, A.B., #Iweka, R.C., Smith, K.E., and **Linzmeier, B.J.** Lead Concentrations in the Community Gardens of Mobile, Alabama. GSA Connects.

Linzmeier, B.J., Kita, N.K., Kitajima, K. Ankney, M.E., Jacobson, A.D., and Valley, J.W. Advancing in situ microanalytical calcium isotope geochemistry by SIMS. GSA Connects.

*__Linzmeier, B.J.__ Cephalopod ecology of the past and present from stable isotope techniques. Dauphin Island Sea Lab.

*__Linzmeier, B.J.__ Exploring modern and ancient cephalopod ecology with stable isotope geochemistry. George Mason University.

*__Linzmeier, B.J.__ Revealing paleobiology and paleoclimatology with 10 micrometer spots. Western Michigan University.

Murphy, J.M., Severin, K.P., Helser, T.E., Lanzirotti, A., **Linzmeier, B.J.**, Ugari, T.M., Newville, M., Scicchitano, M.R., Spaleta, K.J., Randy J. Brown, R.J., Spicuzza, M.J., Hertz, E., Rogers, M., Moss, J.H., Howard, K.G., Gagnon, A.C., and Valley, J.W. Reconstructing migratory histories and age of Chinook salmon from the Yukon River and northern Bering Sea with oxygen isotopes and trace elements in otoliths. Alaska Marine Science Symposium, Online.

2021 Quinn, D.P., **Linzmeier, B.J.**, Sundell, K., Gehrels, G.E., Goring, S., Marcott, S.A., Meyers, S.R., Peters, S.E., Ross, J., Schmitz, M.D., Singer, B.S. and Williams, J.W. Implementing the Sparrow laboratory data system in multiple subdomains of geochronology and geochemistry. EGU General Assembly, Online.

Linzmeier, B.J., Quinn, D.P., Idzikowski, C., Peters, S.E., Kitajima, K., Kita, N.T., Bonamici, C., and Valley, J.W. Connecting a SIMS laboratory to the Sparrow data system. EarthCube annual meeting, Online.

Quinn, D.P., **Linzmeier, B.J.**, Sundell, K., Gehrels, G.E., Goring, S., Marcott, S.A., Meyers, S.R., Peters, S.E., Ross, J., Schmitz, M.D., Singer, B.S. and Williams, J.W. An extensible interface for geochemical data sharing atop the Sparrow laboratory management system. EarthCube annual meeting, Online.

2020 Quinn, D.P., **Linzmeier, B.J.**, Sundell, K., Bruck, B., Ye, S., Gehrels, G., Goring, S.M., Singer, B.S., Marcott, S., Meyers, S.R., Peters, S.E., Ross, J.I., Schmitz, M., and Williams, J.W., The Sparrow laboratory information management system - a tool for connecting geochemical data to context and community. National GSA conference, Online.

Denny, A.C., **Linzmeier, B.J.**, Kitajima, K., and Cammock, J.N. Integrating Geochemical Data and Raster Images to Produce Highly Detailed Sample Maps in a GIS Environment. National GSA Conference, Online.

#Wang, J., #Kitch, G.D., **Linzmeier, B.J.**, Jacobson, A.D., Sageman, B.B., and Hurtgen, M.T. Calcium isotope variability across ancient candidate ocean acidification events. National GSA conference, Online.

Helser, T., Kastelle, C., Crowell, A., **Linzmeier, B.J.**, Scicchitano, M., and Gagnon, A., A paleoclimate record of costal Gulf of Alaska as revealed through *in situ* $\delta^{18}O$ and $\delta^{13}C$ from SIMS and trace elements from LA-ICPMS of ancient Pacific cod otoliths, World Fisheries Congress, Adelaide, Australia.

Linzmeier, B.J.*, Jacobson, A.D., Sageman, B.B., Hurtgen, M.T., Ankney, M.E. and Landman, N.H. Calibrating cephalopod isotope responses using *Nautilus macromphalus* from New Caledonia. National GSA conference, Phoenix, Arizona.

Linzmeier, B.J.* Of seasons and seashells: paleobiology and paleoclimatology from *in situ* and bulk geochemistry of mollusks, University of Florida, Department of Geology, Gainesville, Florida.

2018 **Linzmeier, B.J.*** Big picture, small spots: *in situ* geochemistry applied to paleobiology and paleoclimatology, University of Illinois, Chicago, Department of Geology, Chicago, Illinois.

Linzmeier, B.J. Refining the interpretation of oxygen isotope variability in cephalopods, National GSA conference, Indianapolis, Indiana.

Linzmeier, B.J.* Big picture, small spots: *in situ* geochemistry applied to paleobiology and paleoclimatology, University of Chicago, Department of Geophysical Sciences, Chicago, Illinois.

Linzmeier, B.J., Ward, P.D., Landman, N.H., Peters, S.E., Kozdon, R., Kitajima, K., Tobin, T.S., and Valley, J.W., 2018, Preservation of daily depth migration behavior in $\delta^{18}O$ of cephalopods, 10th International Symposium, Cephalopods - Present and Past, Fes, Morocco.

Linzmeier, B.J.* Big picture, small spots: *in situ* geochemistry applied to paleobiology and paleoclimatology, Northwestern University, Earth and Planetary Science Seminar, Evanston, Illinois.

2017 and earlier are omitted

Selected Professional Posters (# denotes student co-author)

2023 #Magaud, A., **Linzmeier, B.J.**, Beebe, D.A. Vadose Zone Fate and Mobility of Phosphorous in the Little Lagoon Watershed. Bays & Bayous, Mobile, Alabama.

#Small, D., Waldeck, A. R., Jacobson, A. D., Sageman, B. B., and **Linzmeier, B.J.** Assessing Cretaceous *Pycnodonte* Oysters as Calcium Isotope Archives. AGU, San Franscisco, California.

2022 #Small, D., Waldeck, A., Jacobson, A.D., Sageman, B.B., and **Linzmeier, B.J.** Assessing the variability of stable calcium isotopes in Cretaceous carbonate oyster fossils. AGU, Chicago, Illinois.

Linzmeier, B.J., Bolstadt, K., Landman, N.H., and Kitajima, K. Life History of *Architeuthis dux* revealed by SIMS analysis of carbon and oxygen isotopes in statoliths. GSA Connects, Denver, Colorado.

#Leet, K., Lowenstein, T., **Linzmeier, B.J.**, Renaut, R.W., Owen, B., Valley, J.W., and Cohen, A. Conditions of Formation of Magadi-Type Cherts from SIMS Oxygen Isotope Microanalysis. GSA Connects, Denver, Colorado.

2020 **Linzmeier, B.J.**, High Spatial-resolution Assessment of Diagenesis and Primary Isotopic Variability in Maastrichtian Molluscan Carbonates from Antarctica, Microscopy and Microanalysis, Milwaukee, Wisconsin.

Quinn, D.P., **Linzmeier, B.J.**, Sundell, K.E., #Bruck, B.T., #Ye, S., Gehrels, G.G., Goring, S.M., Marcott, S.A., Meyers, S.R., Peters, S.E., Ross, J.I., Schmitz, M.D., Singer, B.S., Williams, J.W., The Sparrow software interface for linking analytical data and metadata in laboratory archives, Earthcube Annual Meeting.

2019 **Linzmeier, B.J.**, Jacobson, A.D., Sageman, B.B., Hurtgen, M.T., Ankney, M.E., Petersen, S.V., and Tobin, T.S., Calcium Isotope Evidence for Environmental Change Before and Across the K-Pg Extinction, Goldschmidt Conference, Barcelona, Spain.

2018 **Linzmeier, B.J.**, Jacobson, A.D., Sageman, B.B., Hurtgen, M.T., Ankney, M.E., and Landman, N.H., Using sub-fossil and historic *Nautilus* to test potential links between calcium isotope fractionation and pCO_2, National GSA conference, Indianapolis, Indiana.

Linzmeier, B.J., Jacobson, A.D., Sageman, B.B., Hurtgen, M.T., Ankney, M.E., Petersen, S.V., and Tobin, T.S., Linking Deccan volcanism and the bolide impact with Ca isotope stratigraphy from the late Maastrichtian Seymour Island, Antarctica, Goldschmidt Conference, Boston, Massachusetts.

Linzmeier, B.J., #Hong, S., and Zaffos, A., The geographic distribution of ammonite morphotypes, 10th International Symposium, Cephalopods - Present and Past, Fes, Morocco.

2017 and earlier omitted

Research Grants (Total: $426,168)

2023 Longleaf Carbon Capture and Storage Collaborative with Southern States Energy Board with Department of Energy Funding (Local Lead PI) $253,868.00 to USA

Public health impacts of soil lead in Mobile, Alabama with Mobile County Public Health Department (Local Lead PI) $150,000.00 to USA

2022 U. South Alabama Social Justice Research: Covariation of demographics and soil lead distribution in Mobile, Alabama

2021 U. South Alabama Faculty Development: Calibrating microanalysis of calcium isotopes in carbonates

2018 The Paleontological Society: Ecology of ancient and modern squid from statolith geochemistry

2013 GSA: Western Interior Seaway Thermohaline Stratification by Ion Microprobe of Ammonites

2010 SIGMA XI: Seasonality of Epicontinental Seas recorded in epithecal banding of Paleozoic Rugose Coral

2009 BGSU: Taphonomic and Ontogenetic Study of Orthocone Nautiloids of the Pennsylvanian Buckhorn Asphalt of Oklahoma Using Temporal Resolved Data Collected via LA-ICP-OES

BGSU: The Forgotten Mastodon: Preservation and Description of a Historic Proboscidean Discovery from 1884, Wood County, Ohio

Honors/Awards (Total: $125,235)

University of South Alabama

2022 Mortar Board Senior Honor Society - "Top Prof"

University of Wisconsin–Madison

2017 Dissertation Completion Fellowship

O. B. Shelburne Research Assistantship

College of Letters and Sciences Teaching Fellow

2015 University Housing Honored Instructor

2014 Thomas E. Berg Excellence in Teaching, Dept. of Geoscience

2010 NSF Graduate Research Fellowship Program - Honorable Mention

2009 Weeks Research Assistantship

Bowling Green State University

2008 Barry M. Goldwater Memorial Scholarship

Dennis L. Roder Geological Field Experience Scholarship

Mancuso Field Studies Scholarship

M.S. Lougheed Scholarship

2006 **David Jacques Memorial Scholarship**

Toledo Gem and Rockhound Club Scholarship

2005–09 University Professors Scholarship

Heart of Wisconsin Gem and Mineral Society Memorial Scholarship

Research Mentorship

Emily Morgan (BS at U. South Alabama, *expected graduation 2026*)

William Glisson (BS at U. South Alabama, *expected graduation 2026*)

Jacob Fetner (BS at U. South Alabama, *graduated 2025*)

Alyssa Foster (BS at U. South Alabama, *graduated 2025*)

Adéle Magaud (BS at U. South Alabama, *graduated 2023*)

Rachel Iweka (BS at U. South Alabama, *graduated 2023*)

Amberlee Johnson (BS at U. South Alabama, *graduated 2022*)

Carlyn Chrabszcz (BS at Brown University, *graduated 2021*)

Caroline Schuette (BS at Northwestern University, *graduated 2018*)

Sungyoon Hong (BS at UW-Madison, *graduated 2016*)

Daniella Assing (BS at UW-Madison, *graduated 2016*)

Steven Henning (BS at UW-Madison, *graduated 2015*)

Graduate Committees

Jessie McCraw, University of Alabama PhD advisor Thomas Tobin

Teaching Experience

University of South Alabama (*as Professor*)

Invertebrate Paleontology
Sedimentology and Stratigraphy
Field Camp
Introductory Geology
Earth History
Stratigraphy
Sedimentology

Workshops and short courses:

Earthcube Geochronology Sparrow online before GSA 2020
QGIS for microanalysis at DMG-MSA Diffusion short course, RUB, Germany 2018

University of Wisconsin–Madison (*as lecturer*):

Introductory Geology, 300+ students, online

Service and Outreach

2025 Chair: Earth Science Curriculum committee

Member: A&S Community Engagement committee

MéPLE Mentorship Program

2024 Chair: Earth Science Curriculum committee

Member: A&S Community Engagement committee

Volunteer: Prospectors & Developers Association of Canada meeting

2023 Invited talk: The Mobile Rock & Gem Society

Chair: Earth Science Curriculum committee

Chair: Earth Science Diversity Equity and Inclusion committee

Member: A&S Diversity Equity and Inclusion committee

GSA Connects Technical Session Chair (2 sessions)

How to build a digital crust workshop instructor (GSA 2023)

2022 Reviewer: Marine Micropaleontology, Geology, Earth and Planetary Science Letters

Chair: Earth Science Diversity Equity and Inclusion committee

University Goldwater Scholarship committee member

Judge: Regional Science and Engineering Fair

Invited talk: The Mobile Rock & Gem Society

2021 Chair: Earth Science Diversity Equity and Inclusion committee

University Goldwater Scholarship committee member

Reviewer: Geology, Earth and Planetary Science Letters, PLoS ONE, ICES Journal of Marine Science, Nature Communications Earth and Environment, Scientific Reports

Invited talk: Slivka Residential College at Northwestern University

2020 Reviewer: Palaeogeography, Palaeoclimatology, Palaeoecology, Geology, PLoS ONE, Paläontologische Zeitschrift

Conference session chair: National GSA conference, Microscopy & Microanalysis

2019 Reviewer: Geochimica et Cosmochimica Acta

Earth and Learn Symposium - Whitney M. Young Magnet High School, Chicago

NAPC session chair

Internships

Chevron Corporation

2015 *Deep-water oil geology: Covington, Louisiana*

2014 *Unconventional oil geology: Midland, Texas*

Academic Internships

2007 *National Museum of Natural History: Washington D.C.*

2006 *The Mammoth Site: Hot Springs, South Dakota*